CLIPMIND AI

Video Summarization & Key Moments Detection Platform

PROJECT DOCUMENTATION

A Final Project Report

Submitted by

HARINI VADIVEL

Technology Stack: React, FastAPI, Python, PostgreSQL, Whisper, Hugging Face Transformers, KeyBERT, FFmpeg.

CERTIFICATE

This is to certify that the project entitled "**CLIPMIND AI – Video Summarization & Key Moments Detection Platform**" is a bonafide project work of **HARINI VADIVEL** who carried out as part of the internship requirements. The project demonstrates the design and implementation of an AI-powered platform for video transcription, summarization, key moment detection, keyword extraction, analytics, and content management.

Project Mentor: **Shanmukha Priya**

Domain: **Artificial Intelligence**

Internship: **Infosys Virtual Internship**

DECLARATION BY THE INTERN

I hereby declare that the project titled "**CLIPMIND AI – Video Summarization & Key Moments Detection Platform**" is my original work of **HARINI VADIVEL** who carried out for internship purposes. The implementation, documentation, and results presented in this report are based on the development work completed for the project, with external technologies and documentation acknowledged appropriately in the references.

SIGNATURE OF THE INTERN

ACKNOWLEDGEMENT

I thank the almighty GOD, without whom it would not have been possible for me to complete my project.

I express my sincere gratitude to my project mentor and everyone who provided guidance and support throughout the development of ClipMind AI. I also acknowledge the open-source communities and technical documentation that supported the implementation of the web, database, video-processing, speech-recognition, and natural-language-processing components.

HARINI VADIVEL

ABSTRACT

ClipMind AI is an intelligent, AI-powered video summarization and content analysis platform designed to simplify the process of understanding and extracting meaningful information from lengthy video content. The system integrates artificial intelligence, speech-to-text processing, natural language processing, and automated content analysis to generate concise summaries, transcripts, key moments, and analytical insights from uploaded videos. The platform uses FFmpeg for audio extraction and Whisper for accurate speech-to-text transcription, enabling the conversion of spoken content into searchable and structured text. The generated transcripts are processed using AI-based summarization models to produce short and detailed summaries while preserving the important information from the original video. A Key Moment Detection module identifies significant portions of the video using keyword and semantic analysis, allowing users to quickly navigate to important timestamps. The system also provides keyword extraction, confidence analysis, compression ratio, reading time, sentence reduction, and ROUGE-based evaluation metrics to assess the quality and effectiveness of generated summaries. ClipMind AI further incorporates an interactive analytics dashboard that presents video insights such as duration, transcript statistics, keywords, key moments, completion rate, and AI-generated content metrics. Users can access transcripts, summaries, key moments, bookmarks, notes, and learning materials through an intuitive web-based interface. The platform also supports content sharing through classroom functionality, enabling educators to share summaries and learning resources with learners. By combining automated transcription, AI summarization, key moment detection, content analytics, and collaborative learning features, ClipMind AI reduces the time and effort required to consume lengthy video content. The proposed system provides an efficient, scalable, and user-friendly solution for students, educators, content creators, and organizations seeking intelligent video understanding and knowledge extraction.

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1. INTRODUCTION

Video content is widely used in education, business, research, training, entertainment, and communication. However, understanding long videos can require substantial time and effort. ClipMind AI addresses this challenge by automatically converting video content into transcripts, summaries, keywords, important moments, analytics, and reusable notes.

The platform combines multimedia processing, speech recognition, natural language processing, and a modern web application architecture to provide a single workspace for video understanding.

2.PROBLEM STATEMENT

Users often need to watch complete videos to locate specific information. Conventional video players do not automatically provide reliable transcripts, summaries, important timestamps, searchable content, or quantitative content insights. This creates a time-consuming manual process, especially for lectures, meetings, interviews, webinars, and long-form tutorials.

The problem is therefore to develop an intelligent system that can automatically analyze video content and present its most useful information in a concise and searchable form.

3.OBJECTIVES

- ❖ Develop an AI-powered video analysis platform.
- ❖ Generate speech-to-text transcripts automatically.
- ❖ Generate short and detailed AI summaries.
- ❖ Detect important moments with timestamps and confidence values.
- ❖ Extract meaningful keywords from transcript content.

- ❖ Provide transcript search and keyword highlighting.
- ❖ Provide video and AI content analytics.
- ❖ Evaluate summary quality using NLP metrics.
- ❖ Combine transcript, summaries, and key moments into Notes.
- ❖ Support PDF/TXT export, copying, sharing, and bookmarks.
- ❖ Provide role-based administrative management.
- ❖ Prepare the platform for cloud deployment.

4.EXISTING SYSTEM

Traditional video workflows require users to manually watch, pause, rewind, and search through lengthy content. Some platforms provide captions or basic search, but they generally do not combine transcription, summarization, key moment detection, keyword extraction, evaluation, notes, and analytics in one workflow.

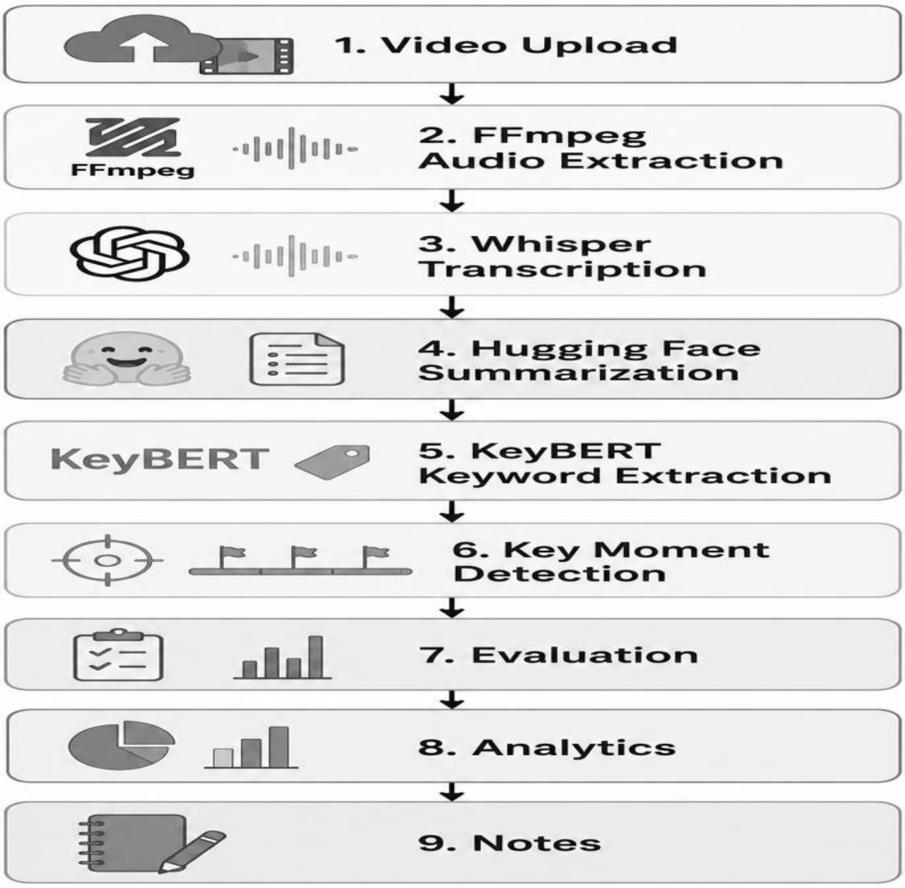
Limitations

- ❖ Manual video navigation.
- ❖ No integrated AI summarization.
- ❖ Difficult identification of important moments.
- ❖ Limited keyword and content analytics.
- ❖ Time-consuming information retrieval.
- ❖ No centralized generated-notes workspace.

5.PROPOSED SYSTEM

ClipMind AI provides an integrated AI-based workflow. A user uploads a video, the system extracts its audio, generates a transcript, analyzes the transcript, produces summaries and keywords, detects key moments, evaluates the summary, and presents the results through an interactive dashboard.

Core workflow:



6.SCOPE

- ❖ Educational lectures and recorded classes.
- ❖ Business meetings and presentations.
- ❖ Interviews and webinars.
- ❖ Training and tutorial videos.
- ❖ Research and academic discussions.
- ❖ Content-creator workflows for identifying highlights.

7.SYSTEM REQUIREMENTS

Hardware Requirements

- ❖ Modern Windows/Linux/macOS development system.
- ❖ Minimum 8 GB RAM recommended for development; more RAM/GPU resources may improve AI processing.
- ❖ Adequate disk space for uploaded videos and AI models.
- ❖ Internet connection for cloud services and package/model downloads when required.

Software Requirements

- ❖ Python 3.10/3.11 environment.
- ❖ Node.js and npm.
- ❖ React with Vite.
- ❖ FastAPI and Uvicorn.
- ❖ PostgreSQL.
- ❖ FFmpeg.
- ❖ Modern web browser.

Functional Requirements

Req. ID	Functional Requirement	Description
FR-01	User Registration	The system shall allow new users to create an account using required details.
FR-02	User Login	The system shall authenticate users using valid login credentials.
FR-03	Role-Based Access	The system shall provide different access permissions for Admin, Educator, and Learner roles.
FR-04	Video Upload	The system shall allow authorized users to upload video files for processing.
FR-05	Video Management	The system shall allow users to view, manage, and delete their uploaded videos.
FR-06	Audio Extraction	The system shall extract audio from uploaded videos using FFmpeg.
FR-07	Speech-to-Text	The system shall generate a transcript from the video's audio using Whisper.
FR-08	AI Summarization	The system shall generate concise and detailed summaries from the transcript using an AI model.

FR-09	Key Moment Detection	The system shall identify important moments from the video and provide their timestamps, titles, descriptions, and confidence scores.
FR-10	Transcript Viewing	The system shall allow users to view the generated transcript and highlighted keywords.
FR-11	Summary Viewing	The system shall display generated summaries in the video details page.
FR-12	Analytics Dashboard	The system shall provide video analytics such as views, watch time, completion rate, word count, summary length, and key moments.
FR-13	Summary Evaluation	The system shall evaluate generated summaries using metrics such as ROUGE scores, keyword coverage, compression ratio, reading time, and overall quality score.
FR-14	Notes and Bookmarks	The system shall allow users to create and manage notes and bookmarks associated with video content.
FR-15	Learning Materials	The system shall allow educators to share summaries and learning materials with learners.
FR-16	Classroom Management	The system shall allow educators to create and manage classrooms and share learning resources through classroom links.
FR-17	Classroom Access	The system shall allow learners to access shared classroom content while restricting editing and deletion to educators.
FR-18	Content Export	The system shall allow users to download or export generated summaries, transcripts, and notes.
FR-19	Admin User Management	The system shall allow administrators to view and manage registered users.
FR-20	Error Handling	The system shall display appropriate error messages when uploads, processing, authentication, or AI services fail.

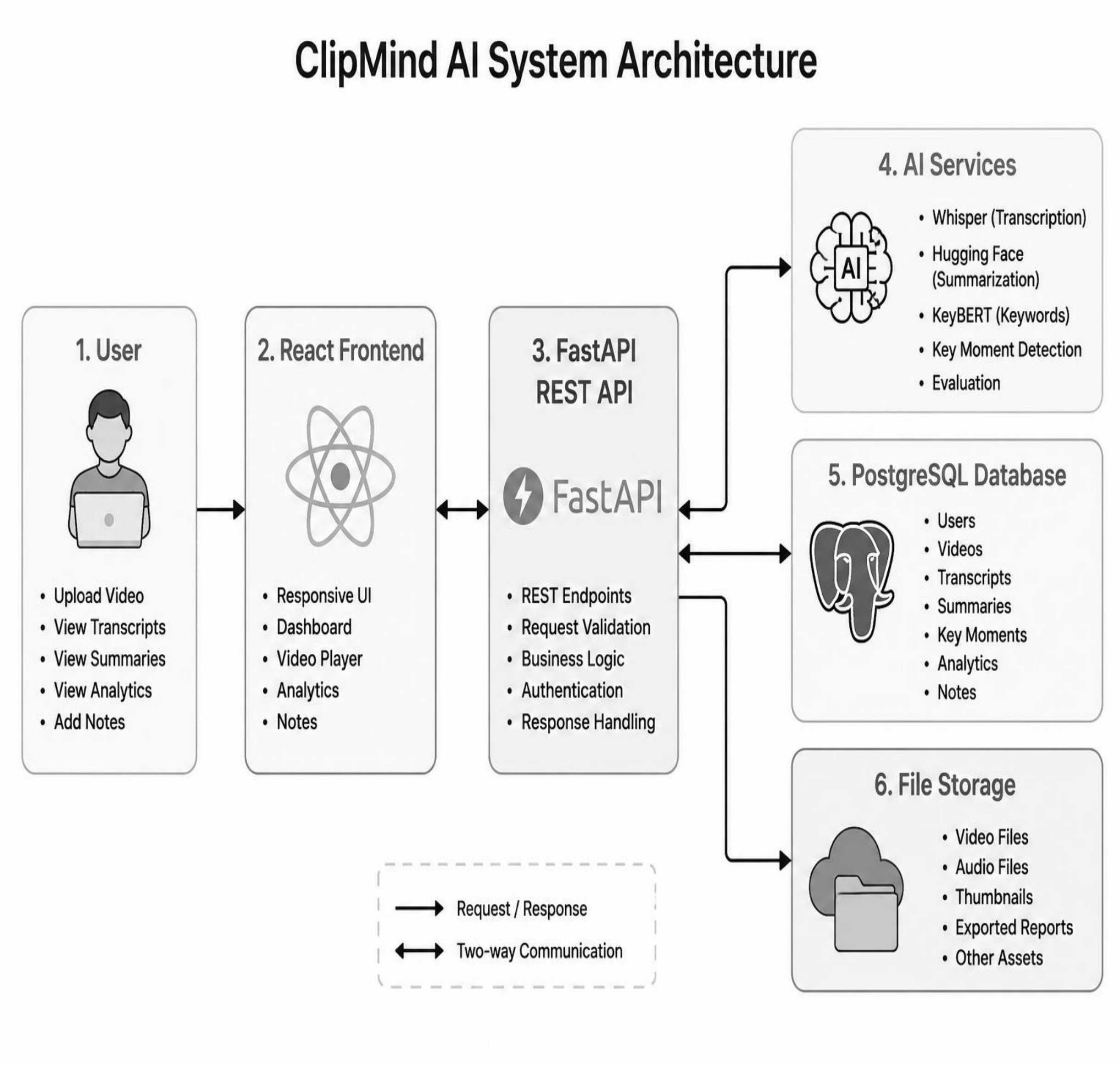
8.TECHNOLOGY STACK

Layer	Technology
Frontend	React, Vite, Tailwind CSS, React Router DOM
Backend	FastAPI, Python, Uvicorn
Database	PostgreSQL, SQLAlchemy
Speech Recognition	Whisper
Summarization	Hugging Face Transformers; BART/T5 variants
Keyword Extraction	KeyBERT with sentence-transformer embeddings
Video Processing	FFmpeg
API	REST
Containerization	Docker
Cloud	AWS / other cloud environments

9.SYSTEM ARCHITECTURE

ClipMind AI follows a layered architecture consisting of a responsive client, REST API backend, AI processing services, database, and media storage. The React frontend communicates with FastAPI. FastAPI coordinates authentication, video processing, AI services, analytics, and database operations.

Architecture:



[Figure 1: System Architecture Diagram.]

10.MODULE DESCRIPTION

10.1 Authentication Module

Provides registration, login, logout, protected routes, user profile management, and role-based access.

10.2 Video Upload and Processing Module

Accepts video files, stores metadata, manages processing status, and coordinates audio extraction.

10.3 Transcript Module

Uses FFmpeg and Whisper to convert video speech into searchable text.

10.4 Summary Module

Uses Hugging Face Transformer models to create short and detailed summaries.

10.5 Key Moments Module

Analyzes transcript content and identifies important sections with timestamps, titles, descriptions, and confidence scores.

10.6 Keyword Module

Uses KeyBERT to identify meaningful keywords and topics.

10.7 Analytics Module

Displays video statistics and AI content statistics including duration, word count, summary length, compression, key moment count, confidence, and keyword metrics.

10.8 Summary Evaluation Module

Calculates ROUGE-1, ROUGE-2, ROUGE-L, compression ratio, keyword coverage, reading time, sentence reduction, and overall quality.

10.9 Notes Module

Combines transcript, summary, and key moments and supports copy, export, and sharing actions.

10.10 Admin Module

Provides authorized administrators with user-management capabilities such as viewing users, changing roles, and deleting users.

10.11 Role-Based Access Control

10.11.1 Content Creator

Features:

- Upload videos
- Manage uploaded videos
- Generate transcripts
- Generate AI summaries
- Detect key moments and highlights
- Download transcripts and summaries
- View content analytics
- Access upload history

Purpose: Create and manage summarized content for audiences.

10.11.2 Learner

Features:

- View uploaded videos
- Read generated summaries
- Access transcripts
- View key moments and timestamps
- Search within transcripts
- Save learning history
- Bookmark summaries and highlights

Purpose: Consume educational and informational content efficiently.

10.11.3 Educator

Features:

- ❖ Upload lecture videos
- ❖ Generate educational summaries
- ❖ Review and edit transcripts
- ❖ Share summaries with students
- ❖ Create learning materials from transcripts
- ❖ Access classroom content analytics
- ❖ Monitor student engagement metrics

Purpose: Transform long educational content into concise learning resources.

10.11.4 Administrator

Features:

- ❖ Manage users and roles
- ❖ Monitor platform activity
- ❖ Manage uploaded content
- ❖ View system analytics
- ❖ Configure platform settings
- ❖ Monitor AI processing jobs
- ❖ Manage storage and resource utilization
- ❖ Access audit logs and reports

Purpose: Maintain platform operations, security, and performance.

11.DATABASE DESIGN

PostgreSQL is used as the primary relational database, with SQLAlchemy providing ORM-based database access.

Users

id,
username,
email,
password/credential data,
role,
created_at

Videos

id,
title,
filename,
duration,
status,
user_id,
created_at

Transcripts

id,
video_id,
content,
language,
created_at

Summaries

id,
video_id,
short_summary,
detailed_summary,
created_at

Key Moments

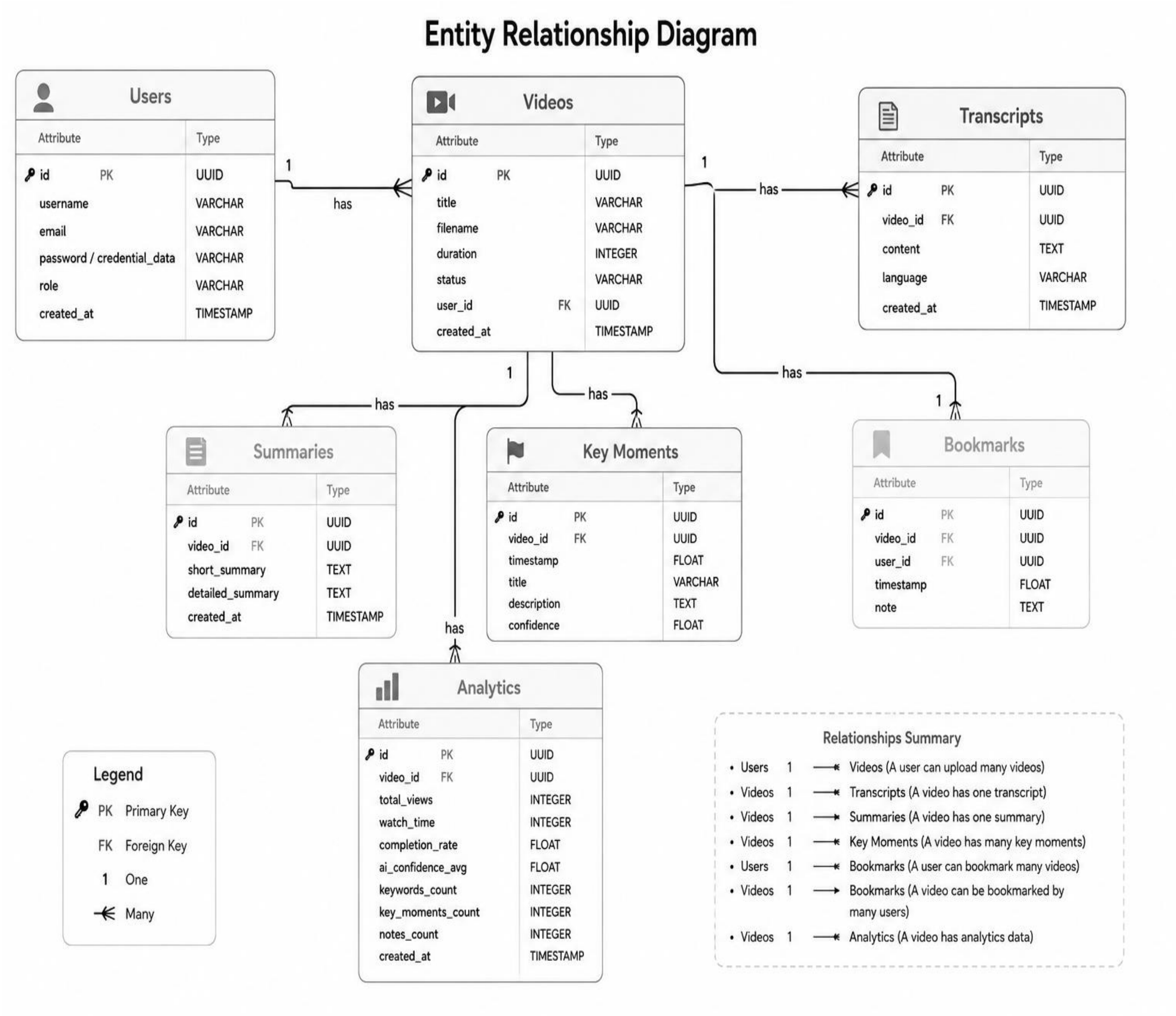
id,
video_id,
timestamp,
title,
description,
confidence

Bookmarks

id,
video_id,
user_id,
timestamp,
note

Analytics

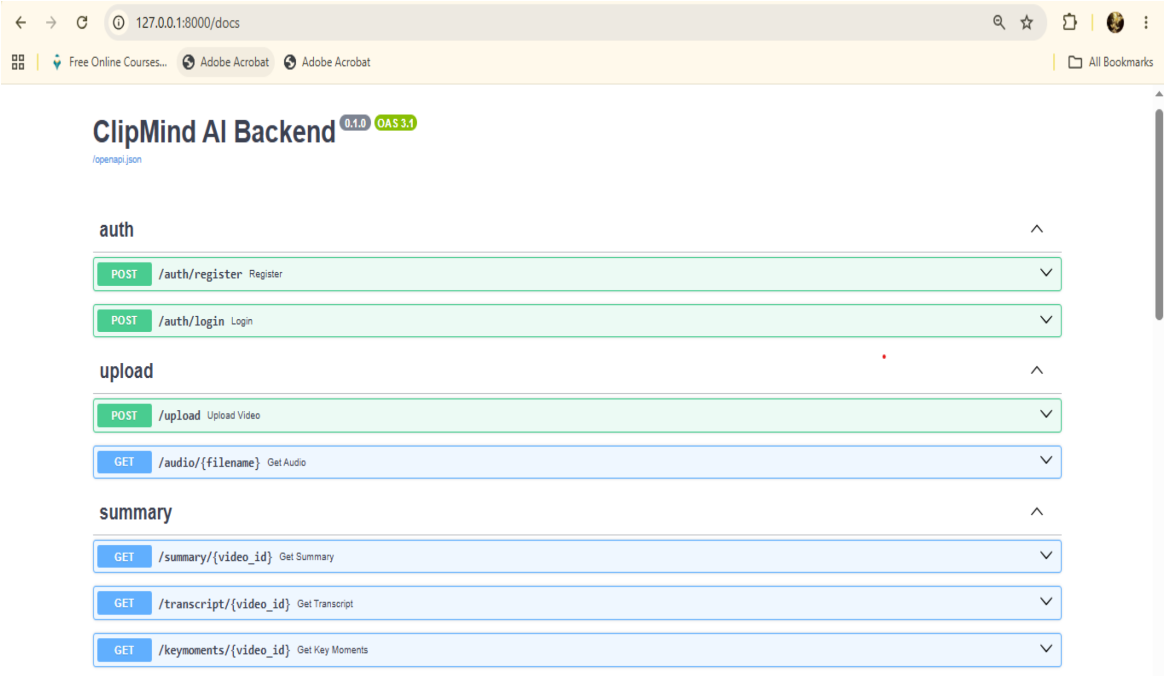
video-related usage and AI content metrics



[Figure 2: Entity Relationship Diagram.]

12.API DESIGN

The FastAPI backend exposes REST endpoints for authentication, videos, transcripts, summaries, key moments, analytics, bookmarks, notes, and administration.



Endpoint Category	Purpose
/api/auth/...	Registration, login, authentication and user access
/api/videos/...	Video upload, listing and management
/api/videos/{id}/transcript	Retrieve generated transcript
/api/videos/{id}/summary	Retrieve or generate summaries
/api/videos/{id}/summary/evaluation	Retrieve summary evaluation metrics
/api/videos/{id}/key-moments	Retrieve detected key moments
/api/videos/{id}/analytics	Retrieve analytics
/api/videos/{id}/bookmarks	Create and manage bookmarks

13.AI/ML IMPLEMENTATION

13.1 Speech Recognition

Whisper is used to convert spoken audio into text. The pipeline extracts audio from video and passes the audio to the speech-recognition model.

13.2 Text Summarization

Hugging Face Transformer models such as BART or T5 variants are used to generate concise and detailed summaries from transcript content.

13.3 Keyword Extraction

KeyBERT identifies keywords and keyphrases based on semantic similarity between document content and candidate phrases.

13.4 Key Moment Detection

Important transcript segments are identified using content relevance, keyword/topic information, and timestamp mapping. Each detected moment is stored with a confidence score.

13.5 Summary Evaluation

The evaluation service calculates overlap-based ROUGE metrics, compression, keyword coverage, reading time, sentence reduction, and an overall quality score.

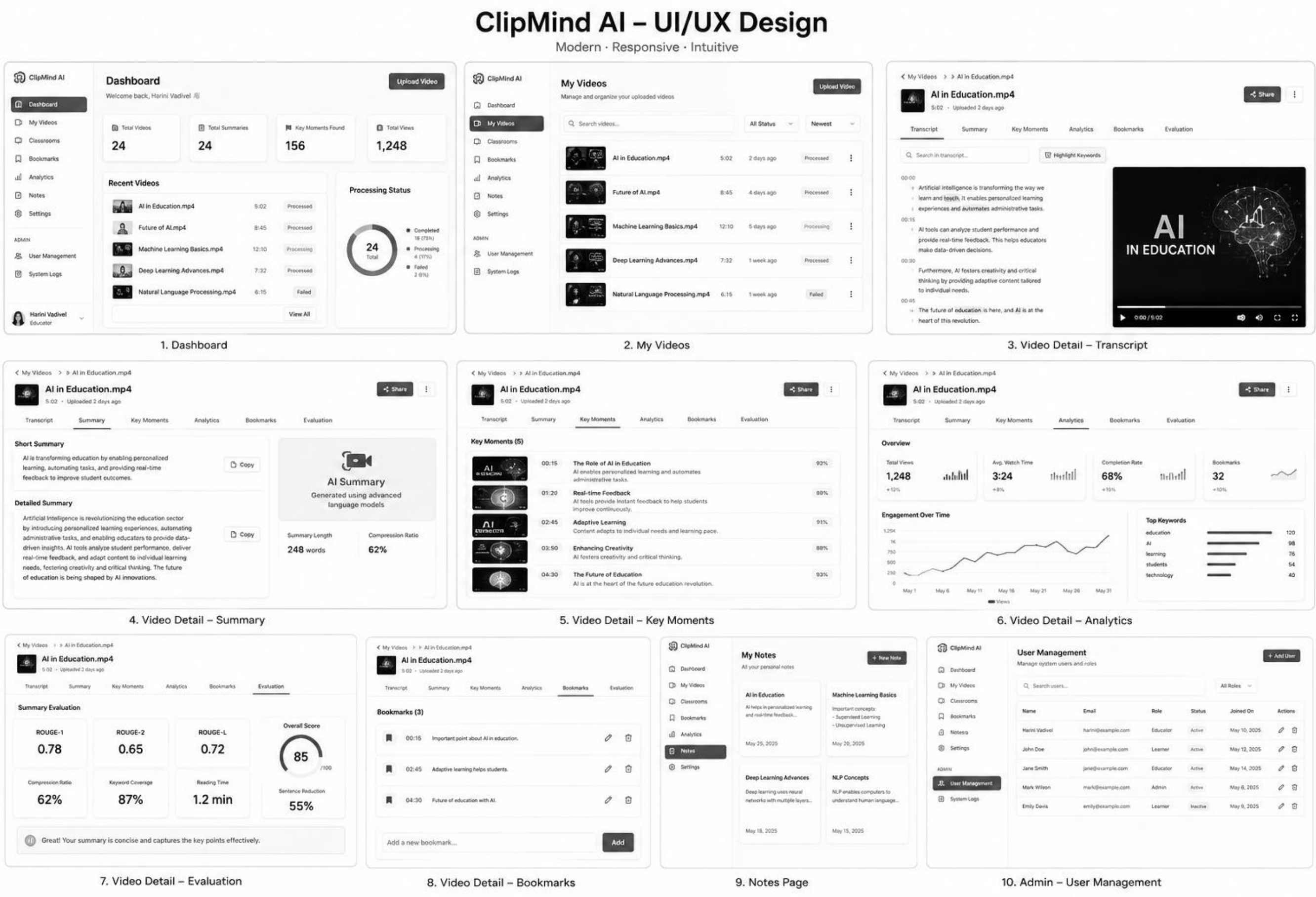
14.UI/UX DESIGN

The frontend uses React and Tailwind CSS to provide a responsive and component-based interface. Navigation is organized around the dashboard and video workspace. The Video Detail page groups Transcript, Summary, Key Moments, Analytics, and Bookmarks into accessible tabs.

- ❖ Responsive layout for desktop, laptop, tablet, and mobile.
- ❖ Clear navigation and dashboard cards.
- ❖ Loading and processing states.
- ❖ Search and keyword highlighting in transcripts.

❖ Interactive analytics and evaluation metrics.

❖ Protected admin interface.



[Figure 3: UI/UX Design.]

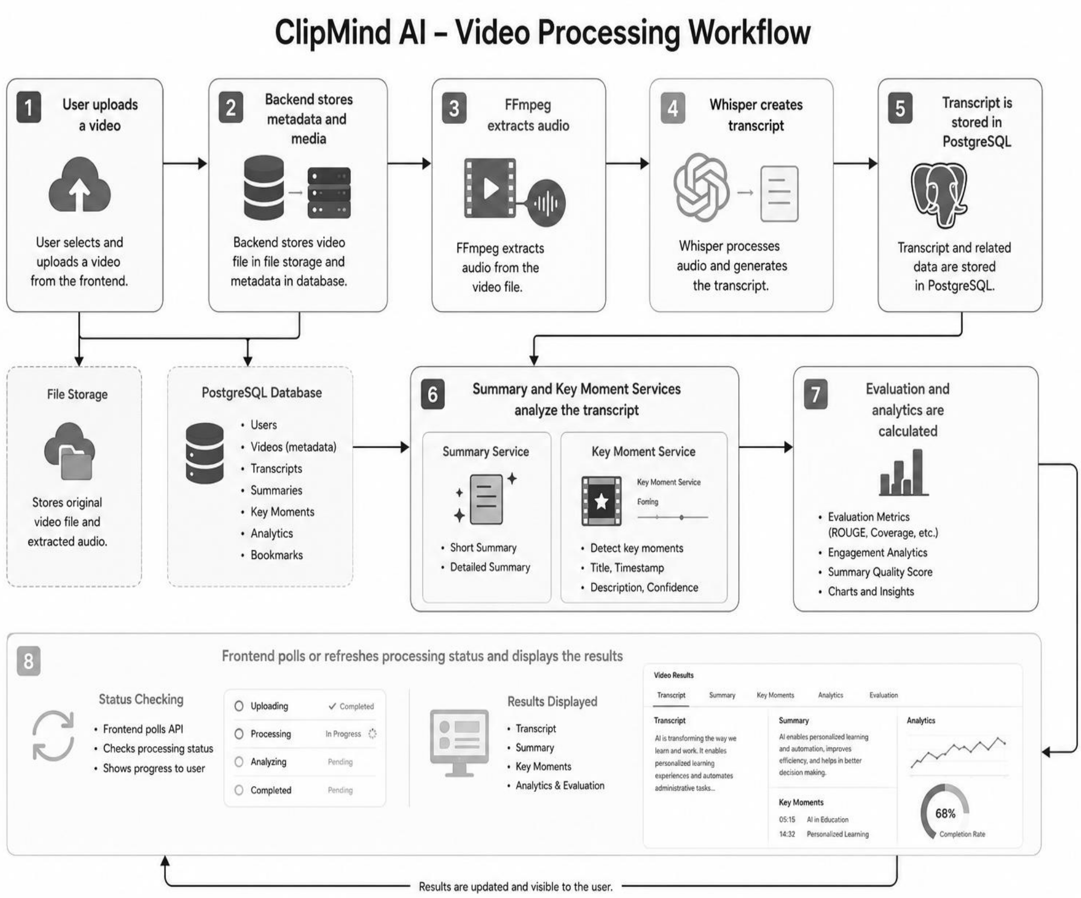
15.IMPLEMENTATION

The frontend is implemented as reusable React components and communicates with the backend through service modules. The backend is organized into API routes, database models, schemas, and service classes. AI processing is separated into specialized services for transcription, summarization, keywords, key moments, and evaluation.

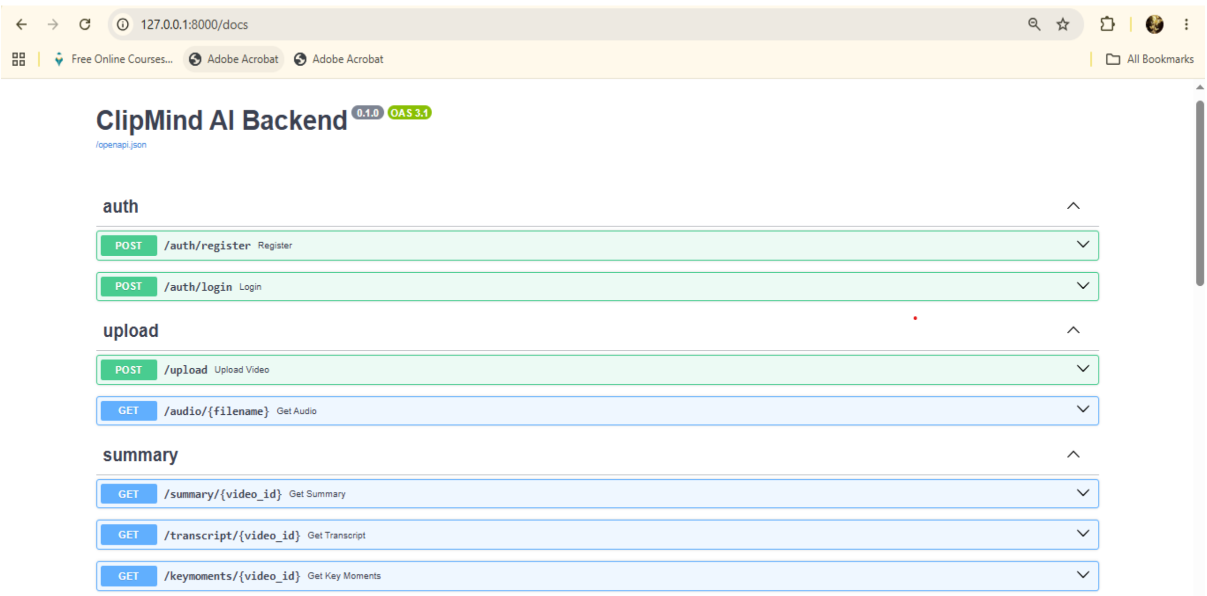
Processing Workflow

1. User uploads a video. 2. Backend stores metadata and media. 3. FFmpeg extracts audio. 4. Whisper creates transcript. 5. Transcript is stored in PostgreSQL. 6. Summary and key moment services analyze the transcript. 7.

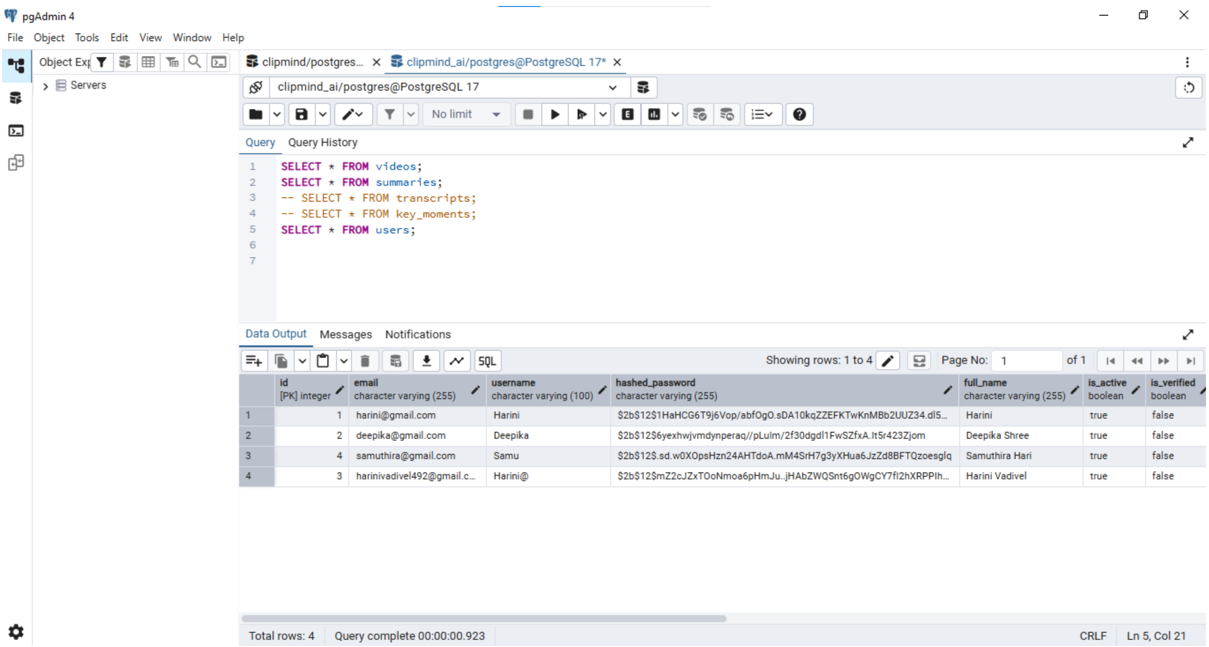
Evaluation and analytics are calculated. 8. Frontend polls or refreshes processing status and displays the results.



[Figure 4: Video Processing Workflow.]



[Figure 5: Backend Server.]



[Figure 6: PostgreSQL pgAdmin4.]

16.TESTING

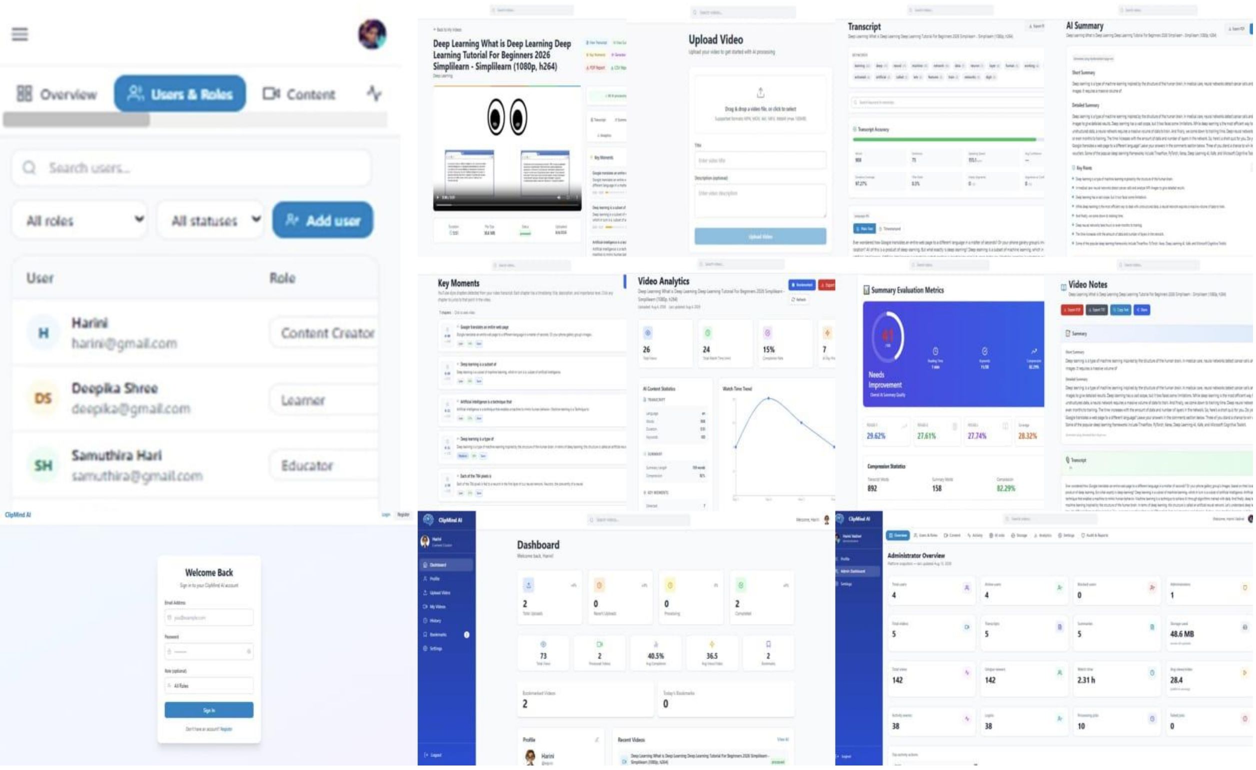
Test Case	Input/Action	Expected Result	Status
User Login	Valid credentials	User enters dashboard	Pass
Video Upload	Supported video	Video is stored and processing begins	Pass
Transcript	Processed video	Transcript is generated and displayed	Pass
Summary	Available transcript	Short and detailed summaries are generated	Pass
Key Moments	Transcript content	Timestamped moments are displayed	Pass
Transcript Search	Search keyword	Matches are highlighted	Pass
Analytics	Processed video	AI/video metrics are displayed	Pass
Admin Route	Normal user opens admin URL	Access is denied	Pass
Invalid API	Unknown video ID	Appropriate error response	Pass

17.RESULTS AND SCREENSHOTS

The implemented platform provides a complete workflow from video upload to AI-generated content and analytics.

- ❖ Video upload and processing interface.
- ❖ Generated transcript.
- ❖ Short and detailed AI summaries.
- ❖ Key moments with timestamps and confidence.
- ❖ Keyword extraction and transcript search.
- ❖ Analytics dashboard.
- ❖ Summary evaluation metrics.
- ❖ Notes page with export options.
- ❖ User and administrator management.
- ❖ Responsive interface.

Recommended screenshot sequence: Login → Dashboard → Video Upload → Video Detail → Transcript → Summary → Key Moments → Analytics → Summary Evaluation → Notes → Admin Dashboard → Mobile/Responsive View.

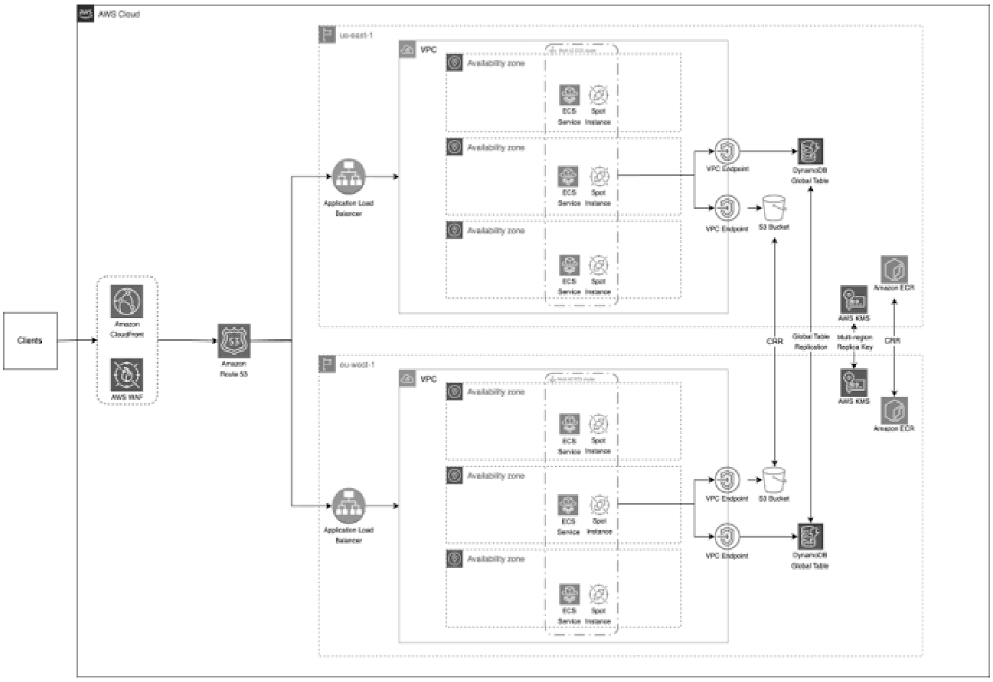


18.DEPLOYMENT

The application can be containerized using AWS EC2. A production architecture can place the React frontend behind a CDN, run FastAPI in a container service such as ECS/Fargate or EC2, use PostgreSQL through Amazon RDS, and store uploaded videos in Amazon S3. An Application Load Balancer, HTTPS certificates, DNS, container registry, and monitoring can be added for a production deployment.

AWS Deployment Components

- ❖ Amazon S3 + CloudFront – frontend and/or media delivery.
- ❖ ECS/Fargate or EC2 – FastAPI backend.
- ❖ Amazon RDS PostgreSQL – database.
- ❖ Amazon S3 – video storage.
- ❖ Amazon ECR – Docker image registry.
- ❖ Application Load Balancer – backend traffic.
- ❖ CloudWatch – monitoring and logs.
- ❖ Route 53 + ACM – DNS and HTTPS.



[Figure 7: AWS Deployment Architecture.]

19.ADVANTAGES

- Reduces time required to understand long videos.
- Automates transcription and summarization.
- Provides important timestamps for quick navigation.
- Makes transcripts searchable.
- Provides quantitative AI content analytics.
- Combines generated content into Notes.
- Supports role-based administration.
- Designed for responsive web access.
- Modular architecture supports future expansion.

20.LIMITATIONS

- Long videos can require significant processing time.
- Speech recognition accuracy can decrease with noise or unclear speech.
- Multiple speakers can reduce transcript quality without dedicated speaker diarization.
- AI models can require substantial RAM/CPU/GPU resources.
- Large media files require significant storage.
- Cloud AI infrastructure can increase operating cost.

21.FUTURE ENHANCEMENTS

- Multilingual transcription and summarization.
- Speaker identification and diarization.
- Real-time meeting transcription.
- Natural-language search across videos.
- Chat with video using retrieval-augmented generation.
- Automatic highlight clip generation.
- Emotion and sentiment analysis.
- Scalable cloud GPU processing.
- Personalized video recommendations.

22.MILESTONES

Milestone	Weeks	Major Work
Milestone 1	Week 1 & 2	Project Initialization, Design Process & Core Setup
Milestone 2	Week 3 & 4	Transcript Generation & AI Summarization
Milestone 3	Week 5 & 6	Key Moments Detection & Analytics Dashboard
Milestone 4	Week 7 & 8	Testing, Deployment & Documentation

23.CONCLUSION

ClipMind AI demonstrates the practical integration of Artificial Intelligence, Natural Language Processing, multimedia processing, and modern web development into a single video-intelligence platform. The system automates transcription, summarization, keyword extraction, key moment detection, analytics, and content organization. Its modular React-FastAPI-PostgreSQL architecture provides a strong foundation for deployment and future enhancements such as multilingual support, speaker identification, conversational video search, real-time processing, and automatic highlight generation.

24.REFERENCES

- ❖ FastAPI Documentation – https://fastapi.tiangolo.com/
- ❖ React Documentation – https://react.dev/
- ❖ Hugging Face Transformers Documentation – https://huggingface.co/docs/transformers/
- ❖ Whisper – https://github.com/openai/whisper
- ❖ KeyBERT – https://maartengr.github.io/KeyBERT/
- ❖ PostgreSQL Documentation – https://www.postgresql.org/docs/
- ❖ SQLAlchemy Documentation – https://docs.sqlalchemy.org/
- ❖ FFmpeg Documentation – https://ffmpeg.org/documentation.html
- ❖ Docker Documentation – https://docs.docker.com/
- ❖ AWS Documentation – https://docs.aws.amazon.com/

25.APPENDIX

A. Suggested Project Structure

```
ClipMind-AI/
├── frontend/
│   ├── src/
│   ├── components/
│   ├── pages/
│   ├── services/
│   └── context/
├── backend/
│   ├── app/
│   ├── models/
│   ├── schemas/
│   ├── services/
│   └── routes/
├── uploads/
├── requirements.txt
├── package.json
├── Dockerfile
└── docker-compose.yml
```

B. Important Configuration

Keep database credentials, authentication secrets, model/API keys, storage credentials, and other sensitive values in environment variables or a secure secrets manager. Do not commit secrets to source control.

C. Suggested Screenshot Captions

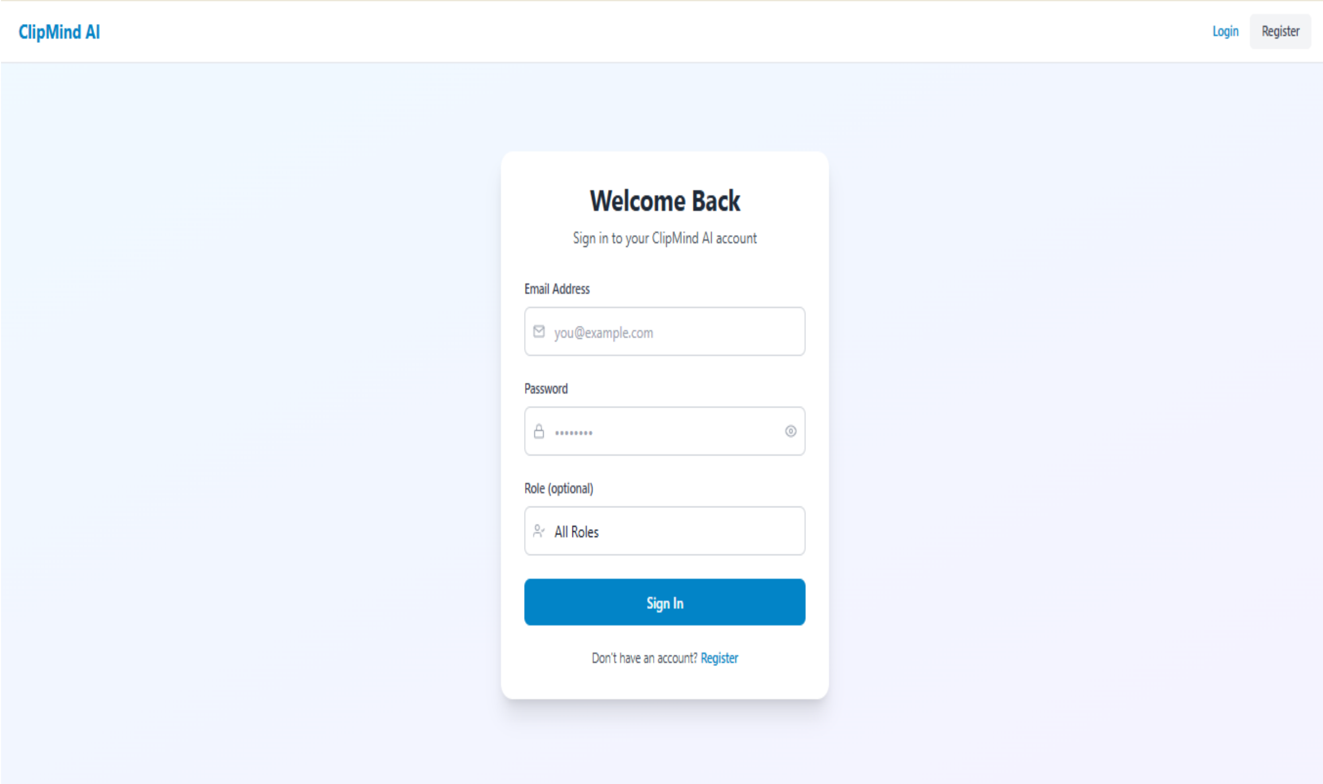


Figure A1 – ClipMind AI Login Page

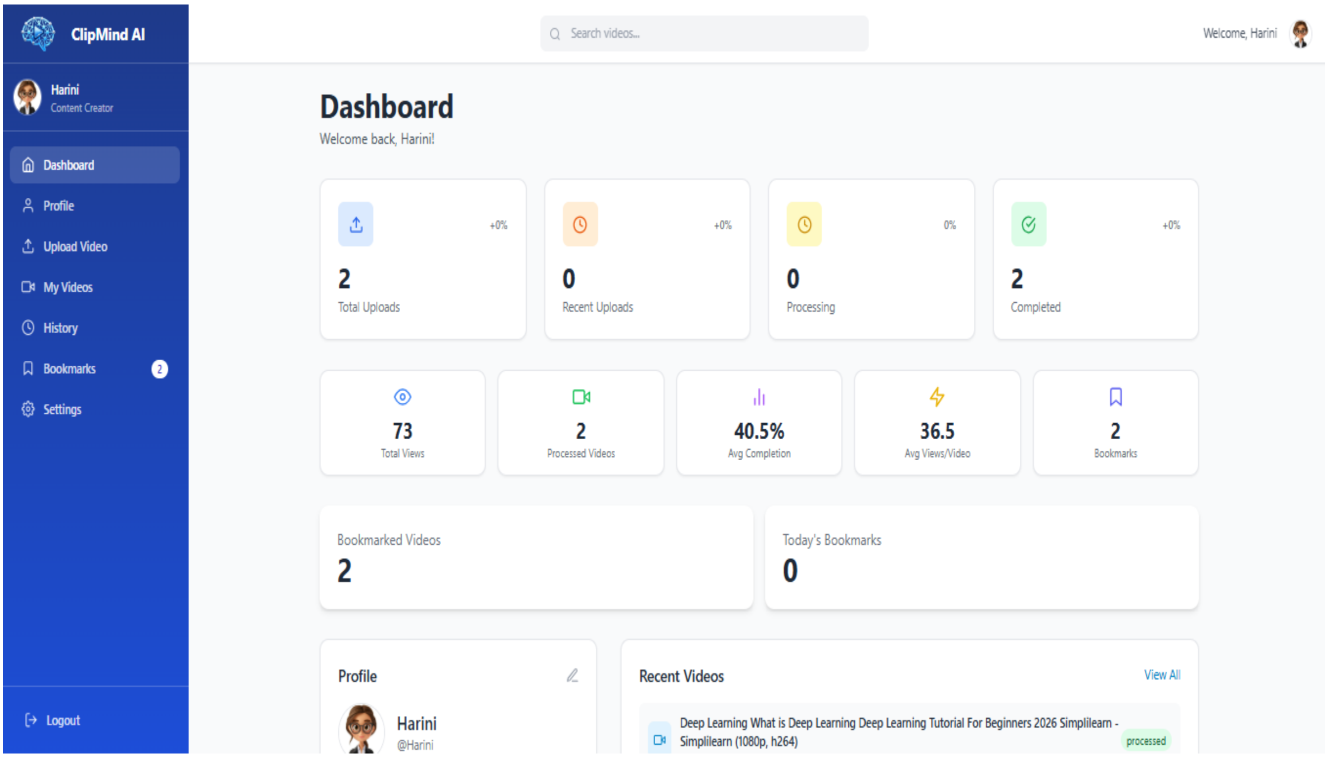


Figure A2 – Main Dashboard

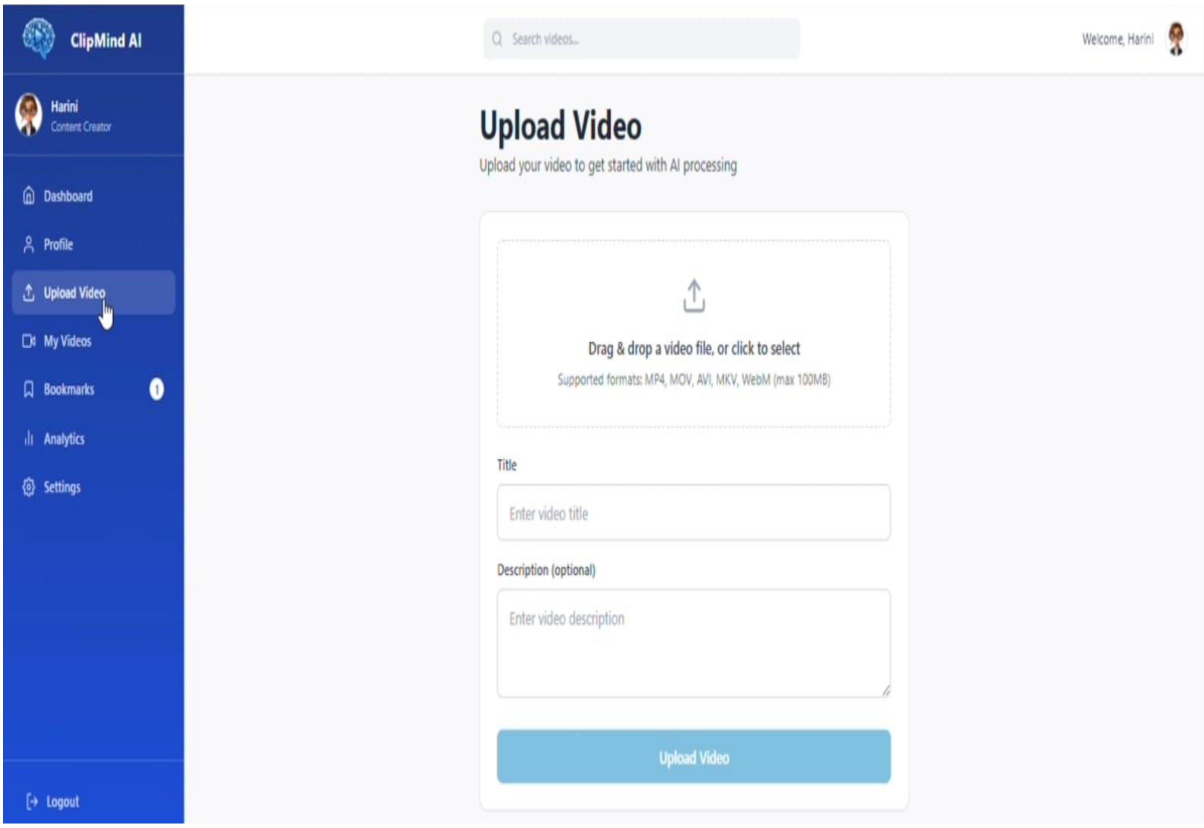


Figure A3 – Upload Video

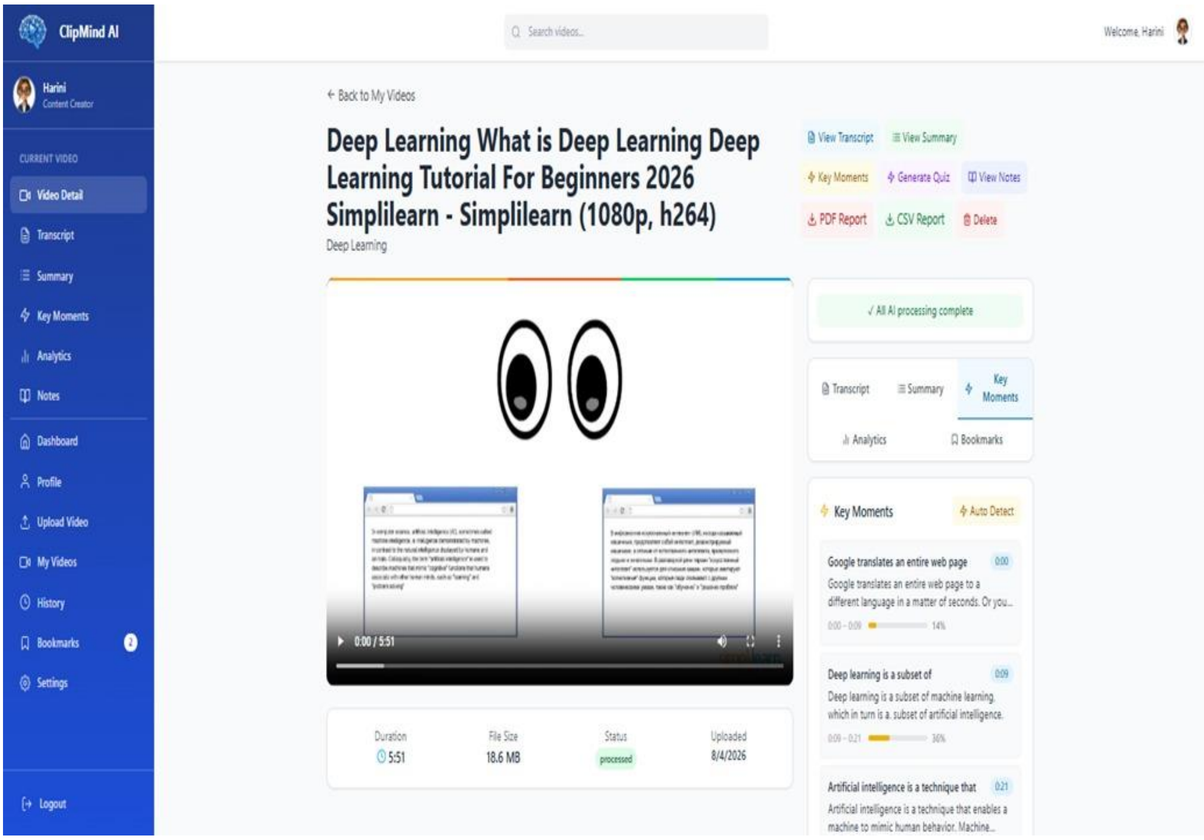


Figure A4 –Video Detail Workspace

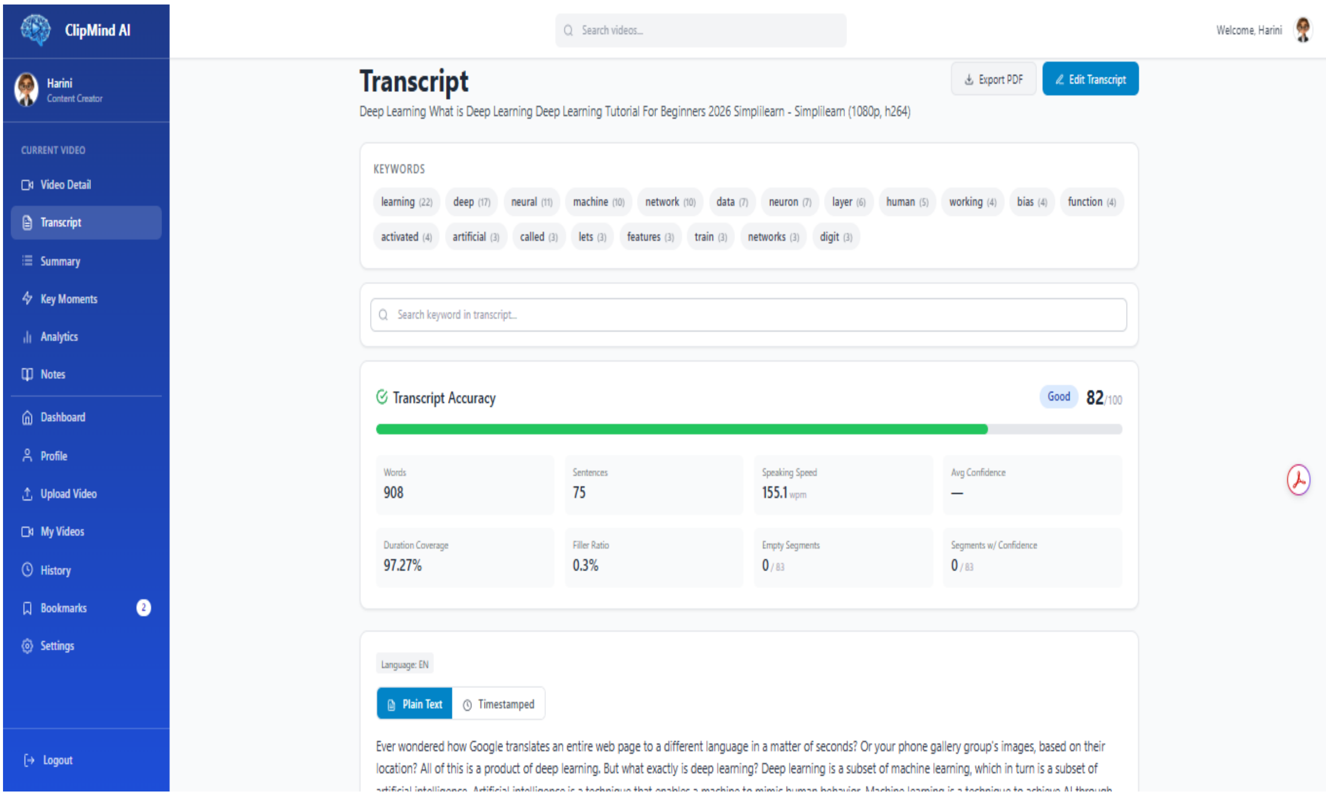


Figure A5 – Generated Transcript

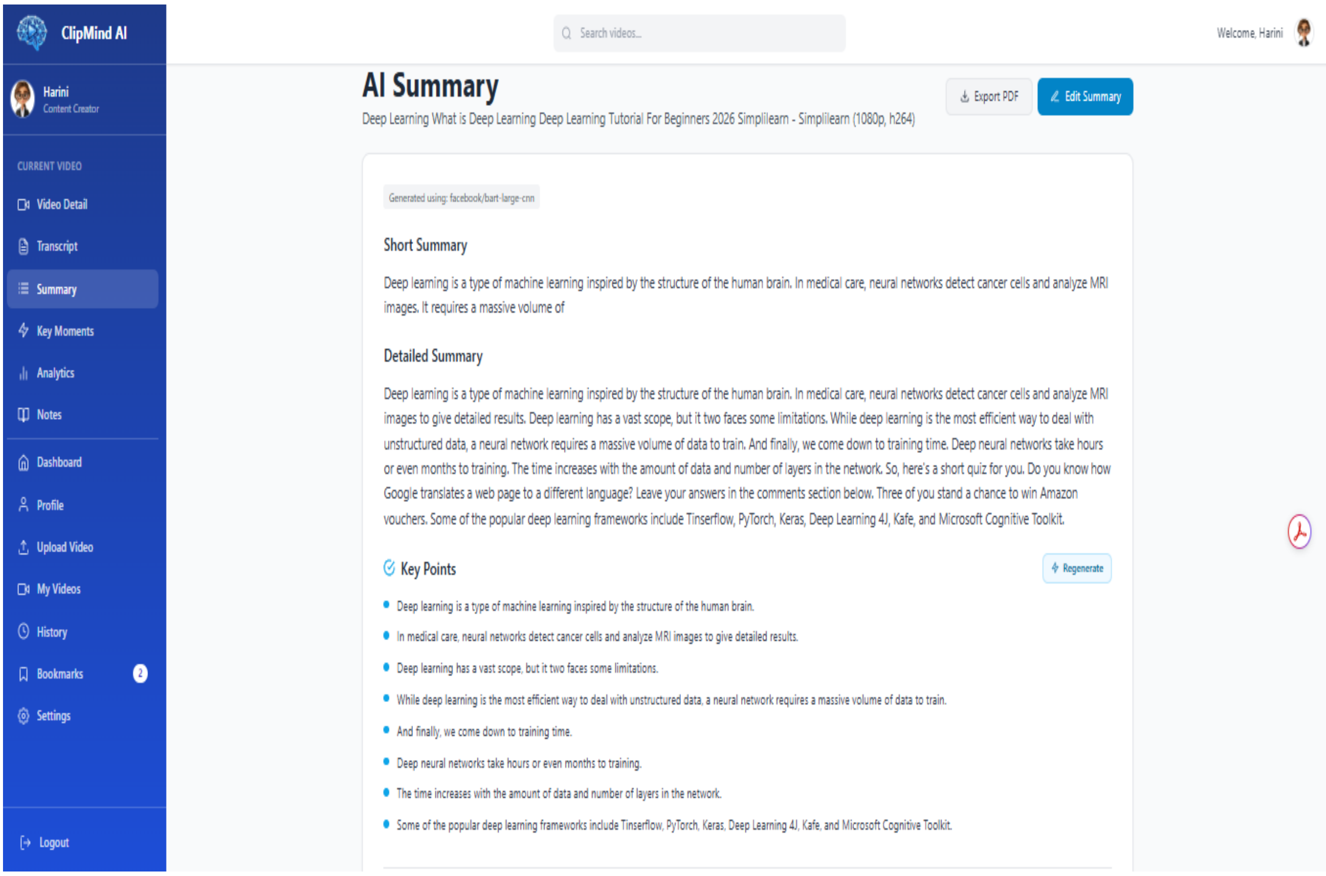


Figure A6 – AI Summary

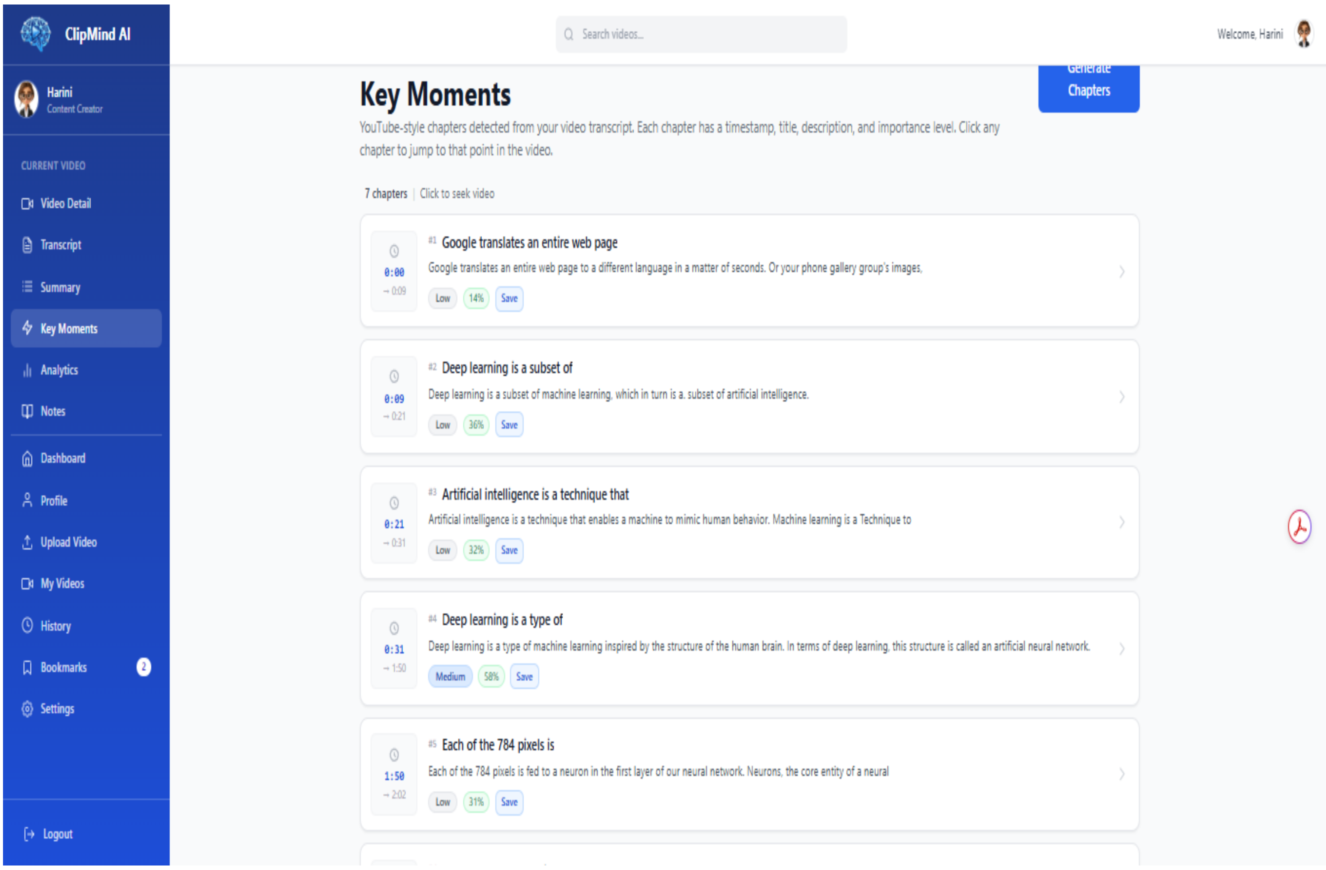


Figure A7 – Key Moments

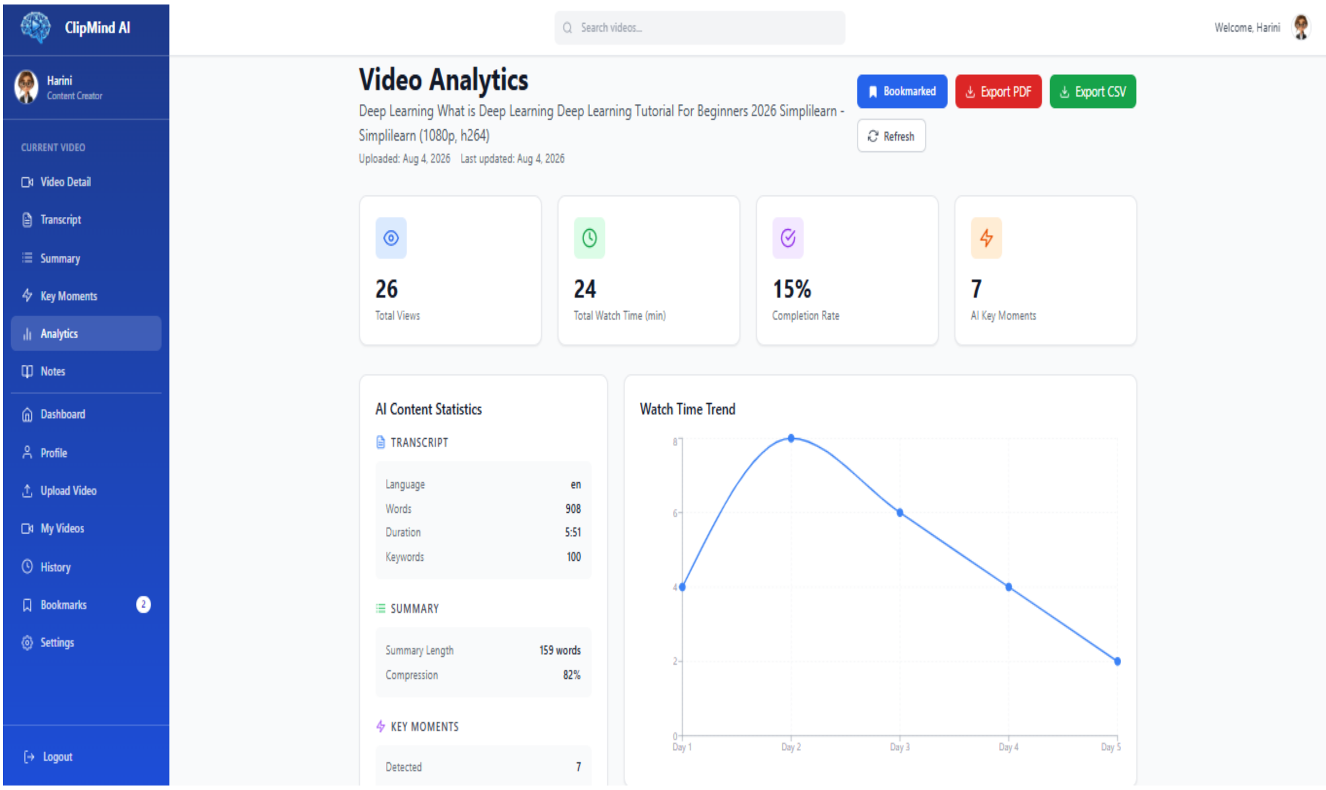


Figure A8 – Analytics Dashboard

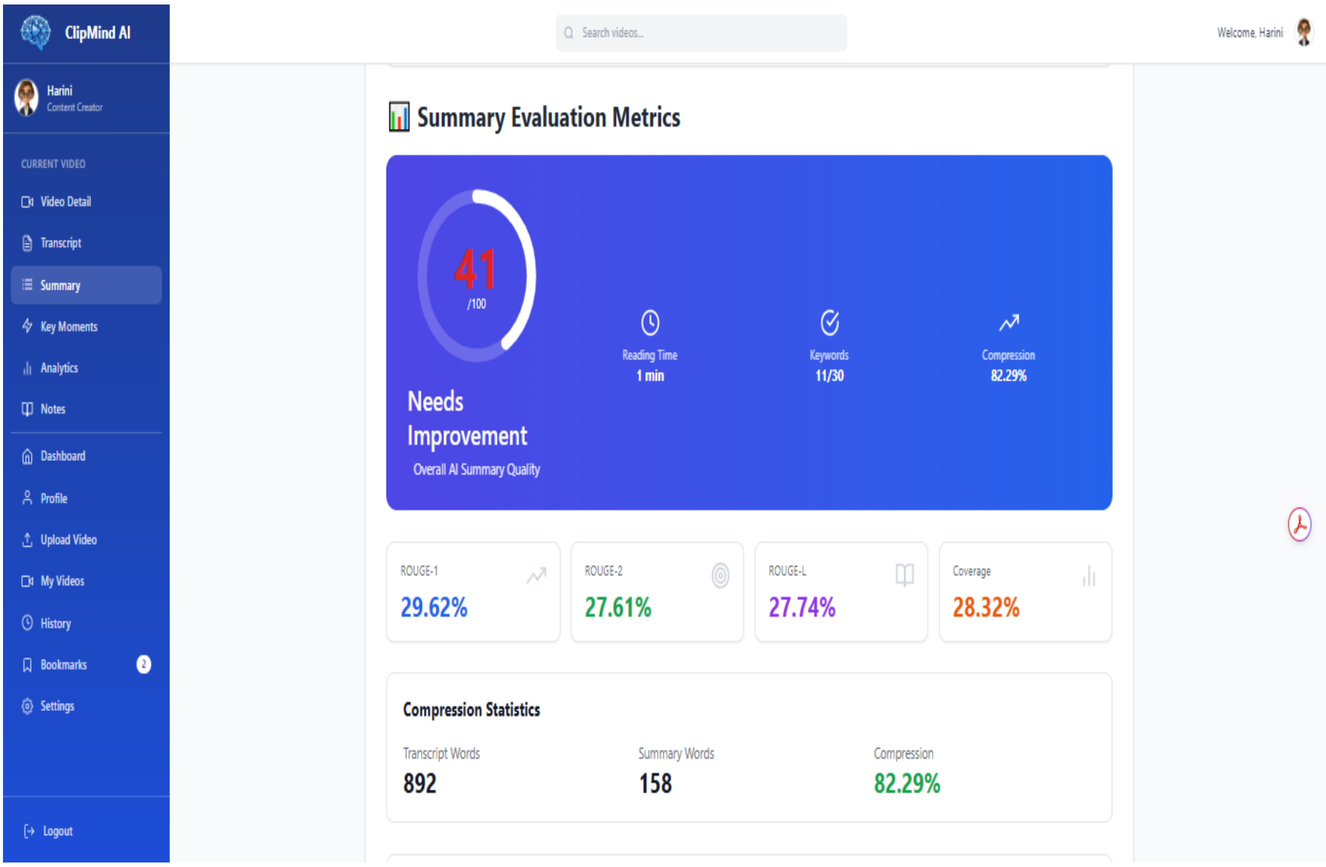


Figure A9 – Summary Evaluation Metrics

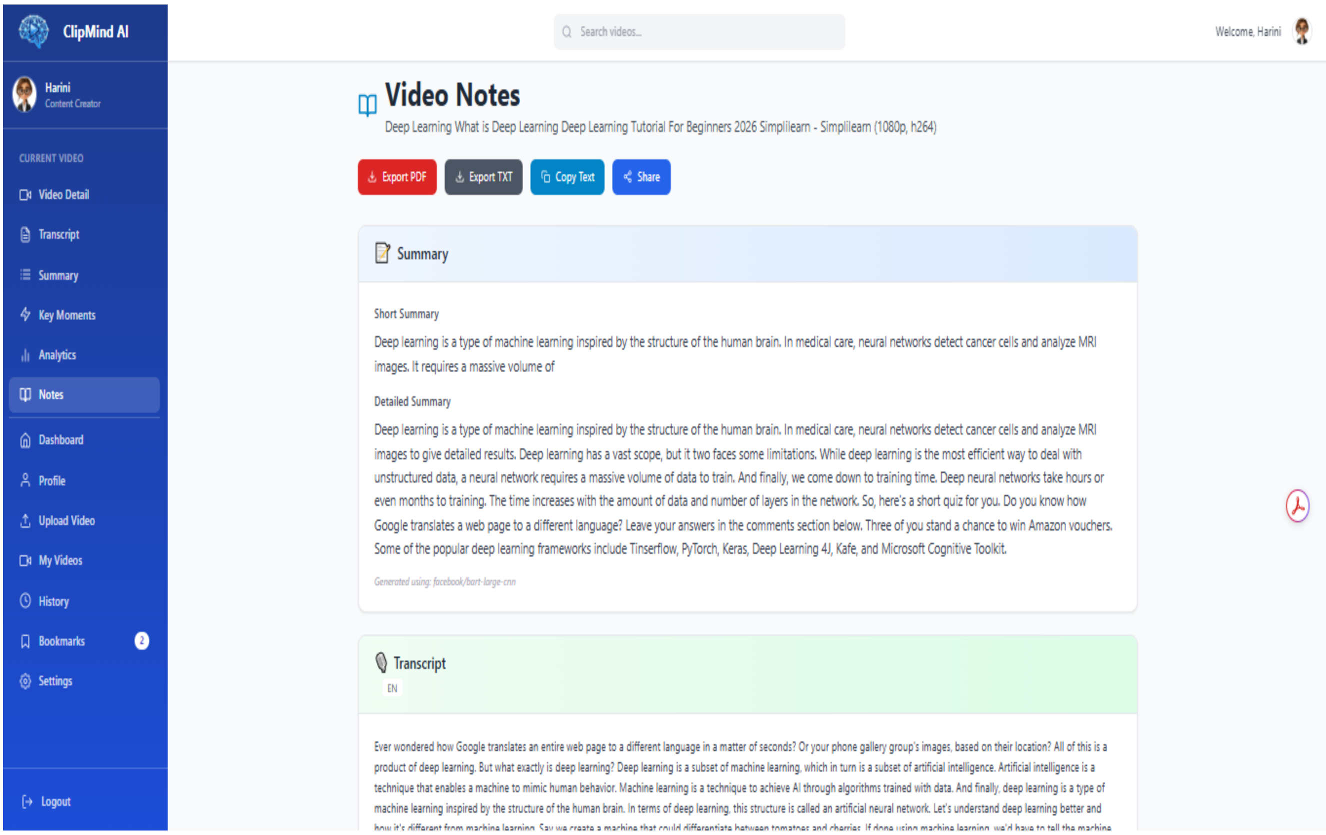


Figure A10 – Notes Page

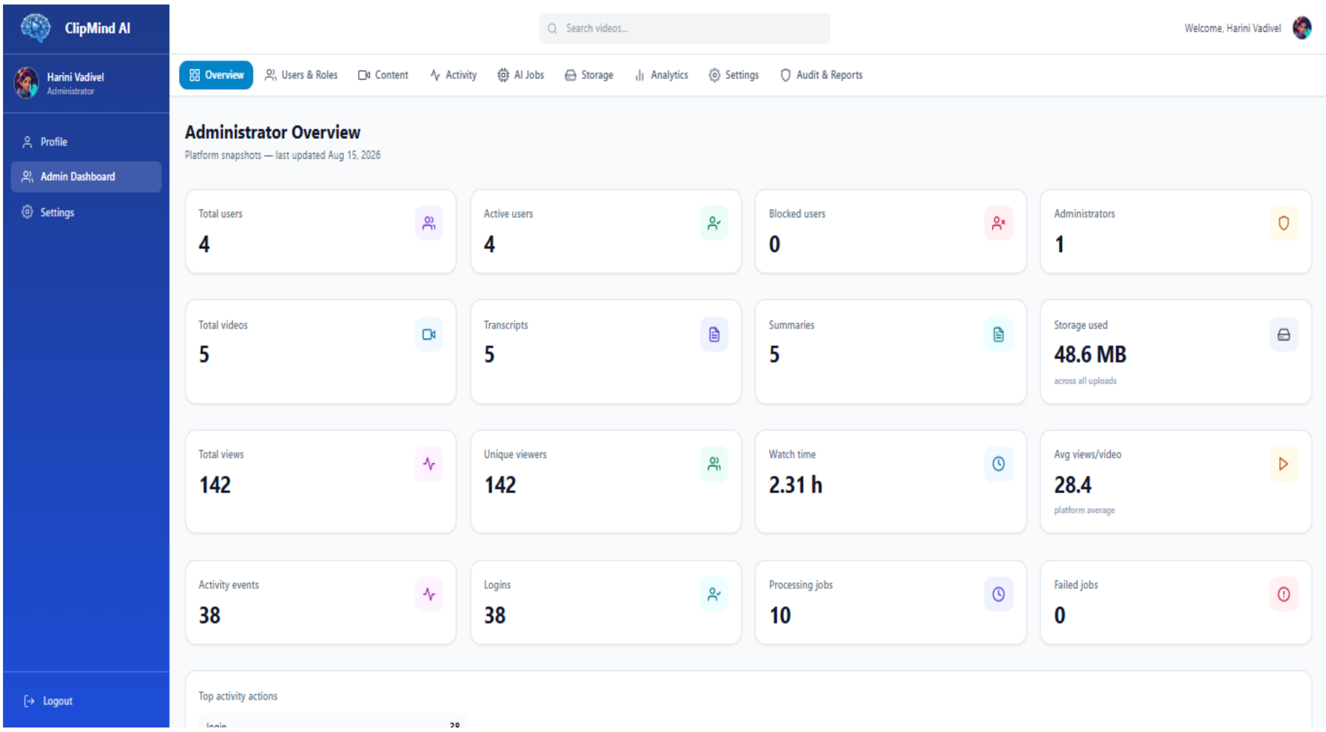


Figure A11 – Admin Dashboard

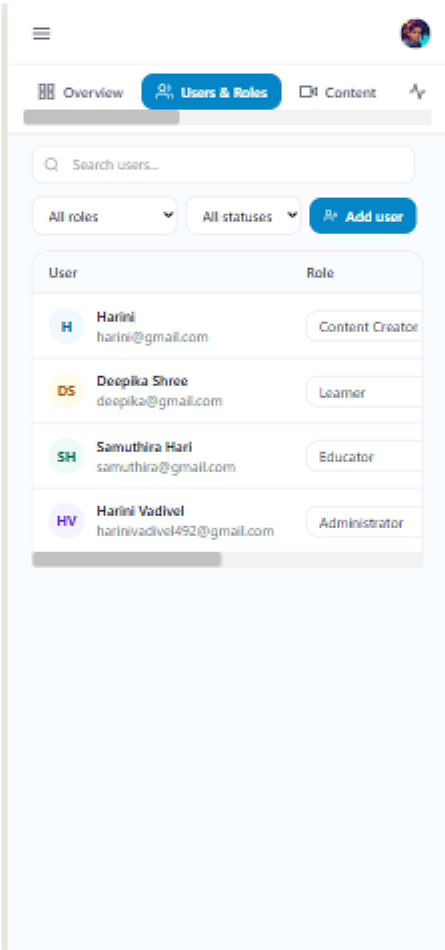


Figure A12 – Responsive Mobile View